determine the speed and direction of the objects in the game, which is important when deciding what action to take.

You can read more about the specifics of DeepMind's DQN by searching for their paper "Human-level control through deep reinforcement learning." One thing to note is that they used a neural network architecture consisting of two convolutional layers followed by two fully connected layers. In our case we used three fully connected layers. It would be a worthwhile experiment to build a model with a convolutional layer, and try training it with Gridworld. One huge advantage of convolutional layers is that they are independent of the size of the input tensor. When we used a fully connected layer, for example, we had to make the first dimension 64—we used a 64 × 164 parameter matrix for the first layer. A convolutional layer, however, can be applied to input data of any length. This would allow you to train a model on a 4 × 4 grid and see if it generalizes enough to be able to play on a 5 × 5 or bigger grid. Go ahead, try it!

3.5. Review

We've covered a lot in this chapter, and once again we've smuggled in a lot of fundamental reinforcement learning concepts. We could have pushed a bunch of academic definitions in your face to start, but we resisted the temptation and decided to get to coding as quickly as possible. Let's review what we've accomplished and fill in a few terminological gaps.

In this chapter we covered a particular RL algorithm called Q-learning. Q-learning has nothing to do with deep learning or neural networks on its own; it is an abstract mathematical construct. Q-learning refers to solving a control task by learning a function called a Q function. You give the Q function a state (e.g., a game state) and it predicts how valuable all the possible actions are that you could take given the input state, and we call these value predictions Q values. You decide what to do with these Q values. You might decide to take the action that corresponds to the highest Q value (a greedy approach), or you might opt for a more sophisticated selection process. As you learned in chapter 2, you have to balance